Research credits and source register

Relation to the paper

This register accompanies *Explicit two-sided bounds for Erdős's minimum-overlap problem* by Cristian Budala. It separates sources used in the research from ingredients in the final proof. The paper itself retains the references needed to follow the argument; this companion records the broader credit without placing an execution history in the paper.

The audit covered61round input manifests,772unique configured paths,548frozen text/Python sources of at most120000bytes, and1241recorded report/supplement files totaling12006605bytes. The report scan produced no additional GitHub repository or arXiv paper leads beyond the source inventory. A bounded script inventory and new primary-source metadata checks supplement that record. This covers the documented research corpus, not unrecorded influences, every transitive software dependency or every model training source. All identified named proof predecessors are cited. Bibliographic checks are not re-proofs of the external authors' certificates.

Direct mathematical lineage

- **Ethan Patrick White:** finite-frequency Fourier/moment relaxation and the basic continuous lower-bound framework.
- **Liam Price:** public mean-bin, positive-part dual certificates and the upstream certificate package. Price's attribution is supported by the problem owner's proof-claim entry; the TeX proof note itself has no author byline.
- **Ruturaj R Raval:** central-bin refinement; his ownCITATION.cff and authorship statement identify him and his contribution. His retained material also credits the Station baseline and Price.
- **Haowen Deng:** weighted-Parseval central certificate. HisCITATION.cff gives family name Deng and given name Haowen. The larger noncentral number is not substituted for a global lower bound.
- **Bautista Machillari Dowguñiec:** subadditivity of the symmetrized overlap and integrated cuts, with exact-arithmetic validation. The byline and title were checked directly in the preserved primary PDF. These mechanisms are inherited, not claimed as new here.
- **Wassily Hoeffding and Colin McDiarmid:** the exponential-moment estimate and bounded-differences framework in the finite rounding argument.
- **Jan Kristian Haugland and P. Swinnerton-Dyer:** step-function upper constructions and the continuous/finite relationship; the Swinnerton-Dyer attribution is made through Haugland. The present finite transfers are proved explicitly and do not assume a limiting-value equivalence.
- **Leo Moser and Paul Erdős:** historical bound and the original problem. The Moser citation was checked in Haugland's primary-paper bibliography; his original paper was not re-audited. The problem owner's bibliography now resolves the original references as four distinct Erdős publications: *Some remarks on number theory*(1955), *Problems and results in additive number theory*(1956), *Some unsolved problems*(1961), and *Some of my forgotten problems in number theory*(1992). Bibliographical records were checked directly at https://www.erdosproblems.com/bibs/Er55, https://www.erdosproblems.com/bibs/Er56, https://www.erdosproblems.com/bibs/Er61, and https://www.erdosproblems.com/bibs/Er92c. The1955and1956items are different works, not competing dates for one title. Their original mathematical texts were not re-proved here.

Historical references in Price's proof-note bibliography also credit **T. S. Motzkin, K. E. Ralston and J. L. Selfridge**, *Minimal overlap under translation*, Bulletin of the American Mathematical Society62(1956),abstract62T-245,p.558, and **S. Świerczkowski**, *On the intersection of a linear set with the translation of its complement*, Colloquium Mathematicum5(1958),185–197. These are indirect historical credits: their original texts were not re-examined and supply no unproved premise here.

Other certificate and research contributions

Sungyoon Kim and Mert Pilanci developed dual-agent convex-relaxation work. **Ben Zanghi** maintains the Bochner/moment-matrix research repository, including the earlier exact512-cell upper certificate used in this workspace. **Nicolas Occis** maintains public lower-certificate and checking reports. Their public profiles

verify those maintainer names; this does not establish individual authorship of every file. Claims of independent checking are attributed as reports, not presented as external human peer review of this paper. The documented Occis exterior/global-bound discrepancy is not imported into our proof.

Stephen Chung, Wenyu Du and William J. Wesley are the authors of the Station paper, which is part of the recorded source lineage and of Raval's acknowledged upstream material. The structured-scaling work listed below was recorded background, not a premise of the final certificate.

Upper construction lineage

The upper witness is incumbent-derived. Round0046starts from MILO's512six-decimal replay values; the displayed sum is255999994/1000000rather than256. Adding six numerator units at coordinate219gives an exactly balanced, box-safe seed. This repaired member is not MILO's original full-precision construction. The Round0047proposal explicitly refines U46; subsequent accepted members pass through U48, U49, the1024-cell refinements U50/U51 and the exact refinements U52–U54. The Round0049proposal reports a sequential-quadratic-programming initialization from the uncoupled U48base. Optimizer history is report-level provenance unless separately reproduced; the accepted coefficient vectors and their objectives are independently checked.

MILO's full author list is **Prithwish Jana, Mononito Goswami, Hao Liu, Xinyu Li, Langlin Huang, Zhehui Huang, Zhishen Huang, Patrick Blöbaum, Anoop Deoras, Purak Jain, Nikos Kanakaris and Sahika Genc**. Their public replay is therefore credited both as source material and as a comparison, not merely mentioned as a competing score. The original full-precision MILO member remains unauthenticated. No latest-record claim follows from the reported-number comparison.

Arena public data and handles such as CodexProLong identify supplied comparison arrays, not verified individual human authors. A separately balanced version of Arena solution2507was checked in the audit; it is not the MILO record. AlphaEvolve and TTT-Discover figures appearing in source tables remain attributed reports unless their original constructions were separately checked. No individual identity is invented for these handles or systems.

Instruments and numerical software

The paper derives its finite Fourier identity, polynomial moments, shared-complement estimate, interval-error bounds and finite transfers. Fubini, Taylor's theorem, convexity, affine interpolation, Markov's inequality and the union bound are standard tools, not claimed inventions. The numerical estimates are stated with their hypotheses and proved or reduced to explicit identities. Moore, Kearfott and Cloud are cited as standard interval-analysis background; this book was added in the editorial revision, not claimed as an original agent reading.

Historical branches include Bochner positivity, Fejér/Parseval constraints, Bernstein-basis positivity, polynomial moment/semidefinite relaxations, Farkas duality and Karush–Kuhn–Tucker conditions. They are not all ingredients of the final theorem. Standard background supplied here includes Bochner's paper, Katznelson's harmonic-analysis text, Lasserre's moment work, Powers and Reznick on positive polynomials, and Nocedal and Wright on optimization. These background references must not be misrepresented as evidence of a specific unlogged consultation or as proof that an unsuccessful search rules out a method.

The standalone final checkers use only Python integer/rational arithmetic. Credit goes to **Guido van Rossum, the Python core contributors and the Python Software Foundation**, whose official history/license record identifies the project's origins and collective contributions. The final checkers do not depend on NumPy, SciPy, SymPy, Arb or a solver.

The exploratory script inventory documents NumPy/SciPy calculations, SLSQP optimization, HiGHS linear programs, SymPy symbolic checks and CVXPY models with SCS/Clarabel selections. Import or solver selection is evidence of source-level tooling, not by itself of an executed successful search. The exact-versus-discovery distinction is retained in the inventory. Arb/python-flint, MPFI and mpmath interval calculations occur in predecessor or checking projects; they are credited as upstream numerical infrastructure, not falsely claimed as dependencies of the present standalone proof. Full bibliographic author lists follow, including the long software contributor lists rather than replacing them by et al.

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Additional upstream software collectives

The **mpmath developers** (https://mpmath.org), **python-flint contributors** (https://python-flint.readthedocs.io), and **FLINT development team** (https://flintlib.org) are credited as upstream software collectives. Their libraries occur in the predecessor/checking ecosystem; they are not dependencies of the two standalone checkers. Individual contributor lists for these transitive projects were not separately audited, so no invented personal author list is supplied.

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Unknown contributor identities and missing original source data are not filled in by inference. AI assistance is disclosed separately and no model is listed as a human author.

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The Erdős Problems database was created by **Thomas Bloom**, as stated in its FAQ; its contributors, including the additional assistance credited to **Terence Tao** on problem36, are acknowledged for the curated bibliographic information. This is not an endorsement or examination of this paper by them.